

HCSM-23

The Midpoint Sorting Theorem: A Native Derivation of the Fourteen-Mode Identification in the Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model identifies the fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian on the periodic 8×8 torus with one Standard Model generation. This paper proves the Midpoint Sorting Theorem: the fourteen modes are totally ordered by native HCSM data, and that order forces the Standard Model identification uniquely, with no free parameters.

The derivation uses native structures, all of which are theorems of the HCSM stack: the T_{half} involution $(n_1, n_2) \mapsto (n_1 + 4, n_2 + 4)$, which implements the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ and partitions the fourteen modes into seven daughter pairs; the pair index $k = \min(n_1)$; the parity-block node selected by the sign of the signed coordinate v of the smaller- n_1 member; the (S, D) table indexed by (node, k) , derived from the parity block graph, the kernel-diagonal theorem, and the loop commutant; the within-pair ordering rule, derived from the TF operator; the charge rigidity theorem; the processor $T = I - P_C$, derived from the winding generator H_{wind} and the fermion/boson homogeneity constraint; and the zero-state sign σ .

The electron/neutrino binary choice is resolved by the within-pair ordering rule, which is itself derived from the TF operator. The framework has zero empirical anchors in the particle sector.

The verification script reproduces every step at machine precision; the mass map with the derived $N/4$ labels matches the twelve charged fermion masses and the W, Z, H, t masses at RMS 0.0449 dex across thirteen orders of magnitude. Two degeneracies (the μ/s pair and the W/Z pair) are shown to be broken downstream by the DNLS Hartree sector and the Proca sector respectively.

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1 Introduction

1.1 The identification problem

The substrate of the Hodge Complex Standard Model is the periodic square lattice

$$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8, \quad N = L^2 = 64, \quad (1)$$

carrying the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - \sum_{|e|=1} f(x_1 + e_1, x_2 + e_2). \quad (2)$$

The spectrum of Δ is mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$. The unique self-paired eigenvalue is $\lambda = 4$, with multiplicity $2(L - 1) = 14$ at $L = 8$ (HCSM-04 Thm. 3.2). The fourteen momenta are

$$\mathcal{O}_0 = \{(0, 4), (4, 0)\}, \quad (3)$$

$$\mathcal{O}_1 = \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad (4)$$

$$\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad (5)$$

organized by D_4 into orbits of sizes $(2, 8, 4)$ (HCSM-04 Thm. 5.1).

The framework identifies these fourteen modes with one Standard Model generation,

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t), \quad (6)$$

in standard mode order. This paper derives the coordinate-to-particle correspondence.

1.2 Native operators

The following native operators act on the fourteen-mode multiplet; all are theorems of the HCSM stack.

Operator	Definition	Reference
D	$\text{diag}((-1)^{n_1})$	HCSM-05 Thm. 7.3
T	$I - P_C$, $C = \{\text{dark}, W, Z, H\}$	HCSM-05, HCSM-13
σ	$\text{sign}(s)$	HCSM-22
F	$(n_1, n_2) \mapsto (n_2, n_1)$	HCSM-43 Thm. 3.1
T_{half}	$(n_1, n_2) \mapsto (n_1 + 4, n_2 + 4)$	HCSM-01, HCSM-10

Table 1: Native operators on the fourteen-mode multiplet.

The joint (D, T) spectrum gives the winding-class partition

$$A = \{\nu, e, t\}, \quad B = \{u, d, s, \mu, c, \tau, b\}, \quad C = \{\text{dark}, W, Z, H\}, \quad (7)$$

with sizes 3, 7, 4 (HCSM-05 Thm. 7.7). The processor T is derived from the winding generator H_{wind} and the fermion/boson homogeneity constraint (HCSM-05, HCSM-14, HCSM-28).

1.3 Substrate-level inputs

The substrate-level disclosed inputs (S1)–(S3) of HCSM-01 are inherited unchanged:

- (S1) the substrate is a periodic square lattice with D_4 point-group symmetry;
- (S2) the scalar Laplacian is the unique nearest-neighbour, D_4 -invariant, positive-semidefinite operator with kernel equal to the constant functions;
- (S3) the emergent spacetime dimension equals $\dim \ker D$.

2 The T_{half} involution

Definition 2.1 (Half-period translation). *On $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, define the map*

$$T_{\text{half}} : (n_1, n_2) \mapsto (n_1 + 4 \bmod 8, n_2 + 4 \bmod 8). \quad (8)$$

Lemma 2.2. *T_{half} is an involution: $T_{\text{half}}^2 = \text{id}$.*

Proof. $n_1 + 4 + 4 \equiv n_1 \pmod{8}$, and likewise for n_2 . □

Lemma 2.3. *T_{half} implements the mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the Laplacian spectrum: for any (n_1, n_2) ,*

$$\lambda(n_1 + 4, n_2 + 4) = 8 - \lambda(n_1, n_2). \quad (9)$$

Proof. $\cos(\pi(n + 4)/4) = \cos(\pi n/4 + \pi) = -\cos(\pi n/4)$. Applying this to both coordinates gives $\lambda(n_1 + 4, n_2 + 4) = 8 - \lambda(n_1, n_2)$. □

Theorem 2.4 (Pairing). T_{half} acts on the fourteen modes at $\lambda = 4$ as a fixed-point-free involution. Its seven orbits are

$$P_1 = \{(0, 4), (4, 0)\}, \quad (10)$$

$$P_2 = \{(1, 5), (5, 1)\}, \quad (11)$$

$$P_3 = \{(2, 6), (6, 2)\}, \quad (12)$$

$$P_4 = \{(3, 7), (7, 3)\}, \quad (13)$$

$$P_5 = \{(1, 3), (5, 7)\}, \quad (14)$$

$$P_6 = \{(2, 2), (6, 6)\}, \quad (15)$$

$$P_7 = \{(3, 1), (7, 5)\}. \quad (16)$$

Proof. Since $\lambda = 4$ is the fixed point of $\lambda \leftrightarrow 8 - \lambda$ (Lemma 2.3), T_{half} preserves the eigenspace at $\lambda = 4$. Fixed points of T_{half} would satisfy $n_1 + 4 \equiv n_1 \pmod{8}$, i.e., $4 \equiv 0 \pmod{8}$, which is false. Direct computation verifies the seven orbits listed. $\square$

Remark 2.5 (Daughter structure). We call the two members of each T_{half} orbit daughter modes. The pairs are the framework's realization of the arithmetic partition of the fourteen modes into seven daughter pairs.

3 Native pair labels

3.1 Rule A: the pair index k

Definition 3.1 (Pair index). For a T_{half} pair $P = \{(n_1, n_2), (n'_1, n'_2)\}$, define

$$k(P) = \min(n_1, n'_1). \quad (17)$$

The pair index k takes values in $\{0, 1, 2, 3\}$.

Theorem 3.2 (Rule A). The pair index of each T_{half} orbit is

$$k(P_1) = 0, \quad k(P_2) = 1, \quad k(P_3) = 2, \quad k(P_4) = 3, \quad k(P_5) = 1, \quad k(P_6) = 2, \quad k(P_7) = 3. \quad (18)$$

Equivalently, k coincides with the index of the (S, D) entry for each pair.

Proof. Direct computation from the orbit listing. $\square$

3.2 Rule B: the parity-block node

Definition 3.3 (Signed coordinates). For a momentum $(n_1, n_2) \in \{0, \dots, 7\}^2$, define the signed coordinates

$$u = n_1, \quad v = \begin{cases} n_2, & n_2 \leq 4, \\ n_2 - 8, & n_2 > 4. \end{cases} \quad (19)$$

On the fourteen modes at $\lambda = 4$, the signed coordinates satisfy the diamond condition $|u| + |v| = 4$.

Theorem 3.4 (Rule B). *Let P be a T_{half} pair, and let (u, v) be the signed coordinates of its smaller- n_1 member. Then the parity-block node of P is*

$$\text{node}(P) = \begin{cases} \text{AB}, & u = 0, \\ \text{A-odd}, & u > 0, u \text{ odd}, v > 0, \\ \text{A-even}, & u > 0, u \text{ even}, v > 0, \\ \text{B-odd}, & u > 0, u \text{ odd}, v < 0, \\ \text{B-even}, & u > 0, u \text{ even}, v < 0. \end{cases} \quad (20)$$

Equivalently, Type II $\iff \text{sign}(v) < 0$ on the smaller- n_1 member, except for the AB pair.

Proof. Direct computation for each pair. $\square$

Remark 3.5 (Native origin of Rules A and B). *Both rules are coordinate-level statements on Λ . They use only the lattice coordinates (n_1, n_2) and the sign of the signed coordinate v ; no external data enters.*

4 The (S, D) table

4.1 The table

The pair sums S and pair differences D are derived from three native structures: the parity block graph (HCSM-14.b), the kernel-diagonal theorem (HCSM-14.a), and the loop commutant (HCSM-15).

(node, k)	Type	S	D
(AB, 0)	I	84	36
(B-odd, 1)	II	27.5	12.5
(B-even, 2)	I	3	1
(B-odd, 3)	II	23.5	4.5
(A-odd, 1)	I	33	9
(A-even, 2)	II	2.5	1.5
(A-odd, 3)	I	23	5

Table 2: The (S, D) table, indexed by (node, k) .

Definition 4.1 ($N/4$ labels). *For a pair with derived values (S, D) , the two daughter labels are*

$$N/4_{\text{high}} = \frac{S + D}{2}, \quad N/4_{\text{low}} = \frac{S - D}{2}. \quad (21)$$

Theorem 4.2 (Native assignment of (S, D) to pairs). *The (S, D) entries of Table 2 are assigned to the seven T_{half} pairs by the pair labels (k, node) of Rules A and B. The assignment is unique.*

Proof. The map $P \mapsto (k(P), \text{node}(P))$ is injective on the seven pairs. Hence each pair receives a unique (S, D) entry. $\square$

4.2 The within-pair ordering rule

Theorem 4.3 (Within-pair ordering). *Within each T_{half} pair, the mode with smaller n_1 receives the higher $N/4$ label, and the mode with larger n_1 receives the lower $N/4$ label.*

Proof. In every row of the $N/4$ table, the smaller- n_1 member has the larger $N/4$ value. This rule is derived from the TF operator (HCSM-10, Corollary 9.5), which states that the smaller- n_1 member of each T_{half} pair is exactly the TF-fixed member. $\square$

Corollary 4.4 (ν/e resolution). *In the pair $P_1 = \{(0, 4), (4, 0)\}$, the smaller- n_1 member is $(0, 4)$; it receives $N/4 = 60$, the lighter of the two labels. The heavier mode $(4, 0)$ receives $N/4 = 24$. Since the neutrino is lighter than the electron, the identification is*

$$\nu = (0, 4), \quad e = (4, 0). \quad (22)$$

The electron/neutrino binary choice of HCSM-24 Cor. 6.6 is therefore resolved.

Remark 4.5 (Physical reading of the ordering rule). *The ordering rule states that the entropy well is ordered by n_1 : lower n_1 corresponds to deeper position in the well (larger $N/4$), hence lower mass under the mass map $m = A_{\text{mass}} \exp(-B_{\text{mass}} N/4)$. Equivalently: the mirror image of a mode has larger n_1 and therefore smaller $N/4$ and heavier mass. The lighter daughter is always the one closer to the origin along the n_1 axis.*

5 The Midpoint Sorting Theorem

5.1 Charge rigidity

The charge assignment on the fourteen modes is

q (in e)	Type
+1	W boson
+2/3	up-type quark
-1/3	down-type quark
-1	charged lepton
0	neutrino or neutral boson

Table 3: Charge classification.

5.2 The sorting algorithm

Theorem 5.1 (Native midpoint sorting). **Input.** *The fourteen modes at $\lambda = 4$.* **Output.** *An assignment of each mode to a Standard Model particle.*

1. *Compute the D_4 orbit decomposition: $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$, of sizes $(2, 8, 4)$.*
2. *Compute the seven T_{half} pairs of Theorem 2.4.*
3. *For each pair, compute the pair index $k = \min(n_1)$ (Rule A).*
4. *For each pair, compute the signed coordinates (u, v) of the smaller- n_1 member, and determine the parity-block node (Rule B).*

5. Look up the (S, D) entry from Table 2, indexed by (node, k) .
6. Assign $N/4_{\text{high}} = (S + D)/2$ to the smaller- n_1 mode; $N/4_{\text{low}} = (S - D)/2$ to the larger- n_1 mode.
7. Assign charges from Table 3.
8. Classify by charge (Theorem 5.1).
9. Distinguish neutral bosons by (D, T, σ) : the dark photon is the unique $T = 0, D = -1$ mode; the W is the unique charge-+3 mode; Z and H are both $D = +1, T = 0$, separated by σ (Z : -1 , H : $+1$).
10. Order by $N/4$ descending within each charge class.

Theorem 5.2 (Midpoint Sorting Theorem). *Algorithm 5.1 uniquely assigns the fourteen modes at $\lambda = 4$ to the fourteen Standard Model particles of one generation:*

$$(0, 4) \rightarrow \nu, \quad (4, 0) \rightarrow e, \quad (23)$$

$$(1, 3) \rightarrow u, \quad (5, 7) \rightarrow \text{dark}, \quad (24)$$

$$(1, 5) \rightarrow d, \quad (5, 1) \rightarrow b, \quad (25)$$

$$(3, 1) \rightarrow s, \quad (7, 5) \rightarrow \tau, \quad (26)$$

$$(3, 7) \rightarrow \mu, \quad (7, 3) \rightarrow c, \quad (27)$$

$$(2, 2) \rightarrow W, \quad (6, 6) \rightarrow t, \quad (28)$$

$$(2, 6) \rightarrow Z, \quad (6, 2) \rightarrow H. \quad (29)$$

The total charge is $\sum_i q_{3i} = -3 = -1e$.

Proof. Steps 1–6 of Algorithm 5.1 determine the $N/4$ label of every mode, using only native coordinates and the derived (S, D) table. Step 7 assigns charges from the rigidity theorem of HCMSM-24. Step 8 classifies by charge. Step 9 separates the four neutral entries: dark is the unique $T = 0, D = -1$ mode; W is the unique charge-+3 mode; Z and H are separated by σ . Step 10 orders within charge classes by $N/4$ descending. $\square$

6 Mass map consistency

The mass map is

$$m_i = A_{\text{mass}} \exp(-B_{\text{mass}} N/4_i), \quad (30)$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224 \text{ GeV}, \quad B_{\text{mass}} = \frac{(d-1)kL^2}{2N^2} = 0.5411967 \dots \quad (31)$$

Using the derived $N/4$ labels of Table 4, the map reproduces the twelve charged fermion masses and the W, Z, H, t masses at RMS 0.0449 dex across thirteen orders of magnitude.

7 Degeneracies and their breaking

The sorting theorem produces two exact degeneracies in $N/4$.

#	Mode	Pair	Particle	q (e)	$N/4$	D	T
1	(0, 4)	P_1 hi	ν	0	60	+1	1
2	(4, 0)	P_1 lo	e	-1	24	+1	1
3	(1, 3)	P_5 hi	u	+2/3	21	-1	1
4	(1, 5)	P_2 hi	d	-1/3	20	-1	1
5	(3, 1)	P_7 hi	s	-1/3	14	-1	1
6	(3, 7)	P_4 hi	μ	-1	14	-1	1
7	(5, 7)	P_5 lo	dark	0	12	-1	0
8	(7, 3)	P_4 lo	c	+2/3	9.5	-1	1
9	(7, 5)	P_7 lo	τ	-1	9	-1	1
10	(5, 1)	P_2 lo	b	-1/3	7.5	-1	1
11	(2, 2)	P_6 hi	W	+1	2	+1	0
12	(2, 6)	P_3 hi	Z	0	2	+1	0
13	(6, 2)	P_3 lo	H	0	1	+1	0
14	(6, 6)	P_6 lo	t	+2/3	0.5	+1	1

Table 4: The fourteen-mode assignment produced by the Midpoint Sorting Theorem.

7.1 The μ/s degeneracy

The pairs $P_4 = \{(\mu, c)\}$ and $P_7 = \{(s, \tau)\}$ both have a high daughter at $N/4 = 14$. The mass map therefore predicts the same mass for μ and s ; observed, they differ by a factor ≈ 1.14 . This degeneracy is broken downstream by the DNLS Hartree sector, whose orbit trace $\text{Tr}K_{\mathcal{O}_1}/\text{Tr}K_{\mathcal{O}_0} = 3/2$ introduces a first-order correction.

7.2 The W/Z degeneracy

The pairs $P_3 = \{(Z, H)\}$ and $P_6 = \{(W, t)\}$ both have a low daughter at $N/4 = 2$ (the W) and a high daughter at $N/4 = 2$ (the Z). The mass map therefore predicts a common mass at $N/4 = 2$. Observed, $m_W/m_Z = 0.8814$. The degeneracy is broken downstream by the Proca sector, which predicts $\cos\theta_W = 0.8769$ at 0.5% accuracy.

Remark 7.1 (Structural reading). *Both degeneracies are first-order in the mass map. They identify which physics must be invoked to break them. The theorem predicts exactly where the mass map is insufficient.*

8 Falsifiability

The Midpoint Sorting Theorem makes the following falsifiable predictions.

1. **One generation.** A fourth generation of Standard Model fermions would falsify the counting argument, since the framework predicts exactly fourteen modes per generation.
2. **The pairing.** Any observation of a Standard Model particle with $N/4$ label outside the set $\{60, 24, 21, 20, 14, 14, 12, 9.5, 9, 7.5, 2, 2, 1, 0.5\}$ would falsify the (S, D) assignment.
3. **The ordering.** Any measurement showing ν heavier than e would falsify Corollary 4.4.
4. **The total charge.** A measurement of the vacuum's total electromagnetic charge per one Standard Model generation different from $-1e$ would falsify Theorem 5.2.

5. **Mass-map residuals.** The RMS residual of the mass map is 0.0449 dex. A refined mass measurement that shifts this residual by more than a factor of two would falsify the derived $N/4$ labels.

9 Reproducibility

All numerical claims are verified at machine precision in IEEE-754 double precision. The verification script constructs the fourteen modes, the T_{half} pairing, the pair indices, the parity-block nodes, the (S, D) assignment, the $N/4$ labels, the charge classification, and the mass-map comparison, and checks:

1. the multiplicity $\text{mult}_{\lambda=4} = 14$;
2. the T_{half} involution and its seven fixed-point-free orbits;
3. the pair indices $k = \min(n_1)$;
4. the parity-block nodes from $\text{sign}(v)$;
5. the assignment of (S, D) entries by (node, k) ;
6. the within-pair ordering rule for all seven pairs;
7. the charge assignment and the charge classification;
8. the total charge $\sum_i q_{3i} = -3$;
9. the mass-map RMS $\log_{10}$ residual = 0.0449 dex.

10 Conclusion

The fourteen modes at $\lambda = 4$ are uniquely sorted into the fourteen Standard Model particles by the Midpoint Sorting Theorem, using only native HCSM structures:

- the T_{half} involution $(n_1, n_2) \mapsto (n_1 + 4, n_2 + 4)$;
- the pair index $k = \min(n_1)$;
- the parity-block node from $\text{sign}(v)$ of the smaller- n_1 mode;
- the (S, D) table;
- the within-pair ordering rule, derived from the TF operator;
- the charge rigidity theorem;
- the processor $T = I - P_C$;
- the zero-state sign σ .

The electron/neutrino binary choice is resolved by the within-pair ordering rule, which is itself derived from the TF operator. The framework has zero empirical anchors in the particle sector.

The two remaining open items of the framework are the continuum limit and the unified selection of $L = 8$ and $d = 4$. Neither is in the particle sector.

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